Measure Theoretic Probability
Lesson 8
Limits of Sequences of Sets: Limit Superior and Limit Inferior

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Outline

1. Motivation
2. Recall: $\limsup$ and $\liminf$ for Numbers
3. Limits of Sequences of Sets
4. Consistency and Examples
5. Complements and Probabilistic Meaning
6. Summary and Exercises

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Sequences of Sets

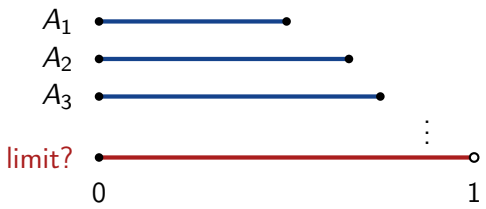
The question

Until now, in expressions such as $\bigcup_n A_n$ and $\bigcap_n A_n$ the indices were only labels: unions and intersections do not depend on the order. Now the order **matters**: A_1 is the first set, A_2 the second, and so on. Can the sequence (A_n) have a **limit**, which we write $\lim_{n\to\infty} A_n$, and is that limit again a set?

A first example in $\Omega = \mathbb{R}$

$$A_n = \left[0,\ 1-\tfrac{1}{n+1}\right]: \quad A_1 = [0, \tfrac{1}{2}],\ A_2 = [0, \tfrac{2}{3}], \ldots$$

The sets are nested and the right endpoint never reaches 1. Intuitively, the limit is $[0, 1)$.



Why a Set-Theoretic Definition Is Needed

Limitation of the metric intuition

In $\mathbb{R}^2$, shrinking circles can be said to "approach" a set because $\mathbb{R}^2$ has a notion of **distance** (a metric space). General sample spaces, such as sets of coin-toss outcomes, have no distance.

Requirement

We need a definition of $\lim_n A_n$ that uses **only unions, intersections and containment**. It must agree with the intuitive answer in the cases where we already feel sure what the limit should be (decreasing and increasing sequences).

Strategy

Mimic the construction of lim sup and lim inf for sequences of real numbers, replacing sup by $\bigcup$ and inf by $\bigcap$.

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Tail Suprema and Tail Infima

Definition

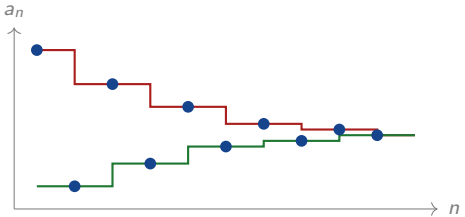
For a real sequence (a_n) and $n \geq 1$,

$$\limsup_{n\to\infty} a_n = \lim_{n\to\infty} \sup_{m\geq n} a_m = \inf_{n\geq 1} \sup_{m\geq n} a_m,$$

$$\liminf_{n\to\infty} a_n = \lim_{n\to\infty} \inf_{m\geq n} a_m = \sup_{n\geq 1} \inf_{m\geq n} a_m.$$

Why the limits exist

Removing terms from the front can only **lower** the tail supremum: $\sup_{m\geq n} a_m$ is non-increasing in n, so its limit exists (possibly $-\infty$) and equals its infimum. Likewise $\inf_{m\geq n} a_m$ is non-decreasing, so its limit is its supremum.



a_n: $\sup_{m\geq n} a_m$ (non-increasing)
$\inf_{m\geq n} a_m$ (non-decreasing).

Limit of a Real Sequence

Definition

If $\limsup_n a_n = \liminf_n a_n$, the common value is called the **limit** $\lim_{n\to\infty} a_n$. If they differ, the limit **does not exist**.

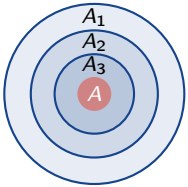
Remarks

- Always $\liminf_n a_n \leq \limsup_n a_n$; the limit exists exactly when the lower and upper "funnels" squeeze together.
- The analogy with sets is only a guide: for sets, sup becomes $\bigcup$, inf becomes $\bigcap$, and the order $\leq$ becomes the containment $\subseteq$.

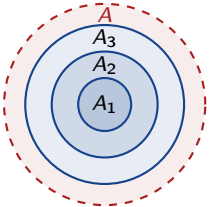
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Monotone Sequences of Sets



$A_1 \supseteq A_2 \supseteq \cdots$, shrinking to $A = \bigcap_n A_n$



$A_1 \subseteq A_2 \subseteq \cdots$, growing to $A = \bigcup_n A_n$

Notation (to be justified by the definition below)

$A_n \downarrow A$ means $A_1 \supseteq A_2 \supseteq \cdots$ and $A = \bigcap_n A_n$. $A_n \uparrow A$ means $A_1 \subseteq A_2 \subseteq \cdots$ and $A = \bigcup_n A_n$.

Tail Unions and Tail Intersections of an Arbitrary Sequence

Definition

For an arbitrary sequence $A_1, A_2, \ldots$ of subsets of Ω, put

$$U_n = \bigcup_{m=n}^{\infty} A_m \qquad \text{and} \qquad I_n = \bigcap_{m=n}^{\infty} A_m.$$

Both families are monotone

- $U_1 \supseteq U_2 \supseteq \cdots$, since U_{n+1} unites **fewer** sets than U_n: so U_n is **decreasing**.
- $I_1 \subseteq I_2 \subseteq \cdots$, since I_{n+1} intersects **fewer** sets than I_n: so I_n is **increasing**.

Consequence

We already know what the limit of a decreasing sequence ($\bigcap$) and of an increasing sequence ($\bigcup$) should be. Applying these two rules to U_n and I_n leads to the definition on the next slide.

Definition: lim sup, lim inf and the Limit of Sets

Limit superior and limit inferior

$$\limsup_{n\to\infty} A_n = \bigcap_{n=1}^{\infty} \bigcup_{m=n}^{\infty} A_m = \bigcap_{n} U_n, \qquad \liminf_{n\to\infty} A_n = \bigcup_{n=1}^{\infty} \bigcap_{m=n}^{\infty} A_m = \bigcup_{n} I_n.$$

Limit

If $\limsup_n A_n = \liminf_n A_n$, this common set is the **limit** $\lim_{n\to\infty} A_n$. Otherwise the limit **does not exist**.

Remarks

- These are definitions, not theorems. $\limsup_n A_n$ and $\liminf_n A_n$ are again **sets**, built from countably many unions and intersections.
- This matches the numeric case: $\bigcap_n \bigcup_{m\geq n}$ is "$\inf_n \sup_{m\geq n}$" and $\bigcup_n \bigcap_{m\geq n}$ is "$\sup_n \inf_{m\geq n}$".

Proposition: $\liminf_n A_n \subseteq \limsup_n A_n$

Proposition

For every sequence (A_n) of subsets of Ω, $\liminf_n A_n \subseteq \limsup_n A_n$.

Proof

Let $\omega \in \liminf_n A_n = \bigcup_n I_n$. Then $\omega \in I_N = \bigcap_{m \geq N} A_m$ for some fixed N, i.e. $\omega \in A_m$ for **all** $m \geq N$. Now fix any $n \geq 1$ and choose $m \geq \max(n, N)$. Then $\omega \in A_m \subseteq U_n$. Since n was arbitrary, $\omega \in \bigcap_n U_n = \limsup_n A_n$. □

Interpretation

The limit exists exactly when the reverse inclusion $\limsup_n A_n \subseteq \liminf_n A_n$ also holds. This parallels $\liminf \leq \limsup$ for numbers.

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Consistency with the Monotone Cases

Decreasing: $A_1 \supseteq A_2 \supseteq \cdots$

- $U_n = \bigcup_{m \geq n} A_m = A_n$ (the largest set is A_n).
- $I_n = \bigcap_{m \geq n} A_m = \bigcap_{m \geq 1} A_m$ for all n (the earlier sets contain A_n and change nothing).

Hence $\limsup = \bigcap_n A_n$ and
$\liminf = \bigcup_n (\bigcap_m A_m) = \bigcap_m A_m$.

Increasing: $A_1 \subseteq A_2 \subseteq \cdots$

- $U_n = \bigcup_{m \geq n} A_m = \bigcup_{m \geq 1} A_m$ for all n.
- $I_n = \bigcap_{m \geq n} A_m = A_n$ (the smallest set is A_n).

Hence $\limsup = \bigcup_m A_m$ and $\liminf = \bigcup_n A_n$.

Conclusion

$\limsup = \liminf$ in both cases, so the limit exists, with $\lim_n A_n = \bigcap_n A_n$ when $A_n \downarrow$ and $\lim_n A_n = \bigcup_n A_n$ when $A_n \uparrow$. The definition recovers our intuition, and justifies the notation $A_n \downarrow A$ and $A_n \uparrow A$.

Example 1: Increasing Intervals

Sequence

$A_n = [0, 1 - \frac{1}{n+1}]$ in $\Omega = \mathbb{R}$: nested increasing.

Computation

- **limsup:** $U_n = \bigcup_{m \geq n} A_m = [0, 1)$ for every n. Intersecting copies of the same set gives $\limsup_n A_n = [0, 1)$.
- **liminf:** $I_n = \bigcap_{m \geq n} A_m = A_n$ by nesting. So $\liminf_n A_n = \bigcup_n A_n = [0, 1)$.

Conclusion

$\limsup_n A_n = \liminf_n A_n = [0, 1)$, so the limit exists, without quotation marks:

$$\lim_{n \to \infty} A_n = [0, 1).$$

Example 2: Two Alternating Sets

Sequence

Let $B, C \subseteq \Omega$ and $A_n = B$ for odd n, $A_n = C$ for even n.

Computation

- Every tail union is $U_n = B \cup C$, independent of n, so $\limsup_n A_n = B \cup C$.
- Every tail intersection is $I_n = B \cap C$, independent of n, so $\liminf_n A_n = B \cap C$.

Conclusion

The limit exists **if and only if** $B \cup C = B \cap C$. Since $B \cap C \subseteq B \subseteq B \cup C$, this forces $B = C = B \cap C$: the limit exists exactly when the sequence is constant.

No monotonicity here

$B \subseteq C$ would not make the sequence increasing: $A_2 = C \not\subseteq A_3 = B$ in general. This example is neither increasing nor decreasing.

Example 2 in a Coin-Toss Space

Setting

Three tosses of a coin: $\Omega = \{HHH, HHT, \ldots, TTT\}$. Take

$$B = \{HHH, HHT, TTH\}, \qquad C = \{TTH, TTT\}.$$

Put $A_n = B$ for odd n and $A_n = C$ for even n.

Results

$$\limsup_n A_n = B \cup C = \{HHH, HHT, TTH, TTT\}, \qquad \liminf_n A_n = B \cap C = \{TTH\}.$$

These sets differ, so $\lim_n A_n$ **does not exist**.

Note

The coin-toss structure plays no role: only the alternation of two fixed sets matters.

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Complements of lim sup and lim inf

Proposition

For every sequence (A_n),

$$\Big(\limsup_n A_n\Big)^c = \liminf_n A_n^c, \qquad \Big(\liminf_n A_n\Big)^c = \limsup_n A_n^c.$$

Proof (De Morgan's laws, applied twice)

$$\Big(\bigcap_n \bigcup_{m\geq n} A_m\Big)^c = \bigcup_n \Big(\bigcup_{m\geq n} A_m\Big)^c = \bigcup_n \bigcap_{m\geq n} A_m^c = \liminf_n A_n^c.$$

The second identity is proved in the same way, or by applying the first to A_n^c and taking complements. □

$\limsup$: "Infinitely Often", $\liminf$: "Eventually"

Proposition

For every sequence (A_n) and $\omega \in \Omega$:

(a) $\omega \in \limsup_n A_n \iff \omega \in A_m$ for **infinitely many** m;

(b) $\omega \in \liminf_n A_n \iff \omega \in A_m$ for **all but finitely many** m.

Proof

(a) $\omega \in \bigcap_n \bigcup_{m \geq n} A_m$ iff for every n there is $m \geq n$ with $\omega \in A_m$. This says exactly that the set of indices $\{m : \omega \in A_m\}$ is unbounded, i.e. infinite.
(b) $\omega \in \bigcup_n \bigcap_{m \geq n} A_m$ iff there is N with $\omega \in A_m$ for all $m \geq N$. This says exactly that the only indices with $\omega \notin A_m$ lie among $1, \ldots, N-1$: they are finitely many. □

Notation and Event Interpretation

Standard notation

$$\{A_n \text{ i.o.}\} = \limsup_n A_n \quad \text{(infinitely often)}, \qquad \{A_n \text{ ev.}\} = \liminf_n A_n \quad \text{(eventually)}.$$

Some texts write $\{A_n \text{ i.o.}\}$ with square or curly brackets, or with none. The i.o. notation is used constantly; "eventually" is rarer.

Events

If the A_n are events of an experiment, then $\{A_n \text{ i.o.}\}$ is the event that **infinitely many** of the A_n occur, and $\{A_n \text{ ev.}\}$ is the event that **all but finitely many** occur.

Consistency checks

- "All but finitely many" implies "infinitely many", which is the inclusion $\liminf \subseteq \limsup$.
- Complements: $\{A_n \text{ i.o.}\}^c = \{A_n^c \text{ ev.}\}$.

Probability of lim sup and lim inf

Setting

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and $A_n \in \mathcal{F}$ for all n.

Measurability

$U_n = \bigcup_{m \geq n} A_m \in \mathcal{F}$ and $I_n = \bigcap_{m \geq n} A_m \in \mathcal{F}$, since a σ-field is closed under countable unions and (by De Morgan) countable intersections. Likewise $\bigcap_n U_n$ and $\bigcup_n I_n$ lie in $\mathcal{F}$. So

$$\limsup_n A_n, \quad \liminf_n A_n \in \mathcal{F},$$

and $\mathbb{P}(\limsup_n A_n)$, $\mathbb{P}(\liminf_n A_n)$, and $\mathbb{P}(\lim_n A_n)$ (when the limit exists) are all **defined**.

Open question

How is $\mathbb{P}\big(\lim_n A_n\big)$ related to $\lim_n \mathbb{P}(A_n)$, the limit of a sequence of real numbers? The next lesson answers this.

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Summary

Key definitions and results

Object	Content
Tail union / intersection	$U_n = \bigcup_{m \geq n} A_m$ (decreasing), $I_n = \bigcap_{m \geq n} A_m$ (increasing)
$\limsup_n A_n$	$\bigcap_n \bigcup_{m \geq n} A_m = \{A_n \text{ i.o.}\}$
$\liminf_n A_n$	$\bigcup_n \bigcap_{m \geq n} A_m = \{A_n \text{ ev.}\}$
Inclusion	$\liminf_n A_n \subseteq \limsup_n A_n$ always
Limit	exists iff $\limsup = \liminf$; common value is $\lim_n A_n$
Monotone sequences	$A_n \downarrow \Rightarrow \lim = \bigcap A_n$; $A_n \uparrow \Rightarrow \lim = \bigcup A_n$
Complements	$(\limsup A_n)^c = \liminf A_n^c$ and vice versa

Exercises

1. Prove that $\sup_{m \geq n} a_m$ is non-increasing in n and $\inf_{m \geq n} a_m$ is non-decreasing in n.
2. Let $A_n = (0, \frac{1}{n}]$ in $\mathbb{R}$. Show the sequence is decreasing and compute $\limsup_n A_n$, $\liminf_n A_n$ and $\lim_n A_n$.
3. Take $\Omega = \{1, 2, 3\}$, $B = \{1, 2\}$, $C = \{2, 3\}$ and alternate A_n between B (odd n) and C (even n). Compute $\limsup_n A_n$ and $\liminf_n A_n$.
4. Show that $\bigcap_n A_n \subseteq \liminf_n A_n$ and $\limsup_n A_n \subseteq \bigcup_n A_n$ for every sequence.

Outlook

Next lessons

- **Continuity of probability:** the relation between $\mathbb{P}(\lim_n A_n)$ and $\lim_n \mathbb{P}(A_n)$. It holds in some situations and fails in others.
- **The Borel–Cantelli lemmas**, which give the probability of $\{A_n \text{ i.o.}\}$.